Sensor HAT v3 Universal — HAT for Raspberry Pi or Seeed XIAO host | SCD41 CO2 | SHT40 T/RH | SGP41 VOC/NOx | VEML7700 light | ICS-43434 mic

PWR_FLAG GND
PWR_FLAG +5V
PWR_FLAG +3V3_PI

Power: Pi 5 V or XIAO USB 5 V via ideal diodes -> +5V -> AP2112K-3.3 LDOs (LDOs on while Pi 3V3 or XIAO 3V3 is present)

C11 10uF 25V
C1 1uF
U5 AP2112K-3.3
C2 1uF
C13 10uF
C3 1uF
U6 AP2112K-3.3
C14 1uF
D3 BAV70
R7 100k

I2C sensors: VEML7700 0x10, SHT40 0x44, SGP41 0x59, SCD41 0x62 (4.7k pull-ups; the Pi adds 1.8k)

C4 10uF
C5 100nF
U1 SCD41-D-R2
U2 SHT40
C6 100nF

XIAO socket J7/J8 with 100R series resistors. NEVER fit a XIAO while the board sits on a Pi

J7 XIAO D0-D6
J8 XIAO 5V-D7
R26 4.7k
R27 4.7k
R31 10k

Raspberry Pi header (female socket, bottom side)

J1 RPI GPIO 2x20

I2S microphone: GPIO18 BCLK, GPIO19 LRCLK, GPIO20 DIN / XIAO D1, D0, D2

C8 100nF
U7 VEML7700-TT
C9 100nF
U3 ICS-43434
R4 100k

JP6 D5 = 1-Wire | radar OUT

ID EEPROM (Pi 3V3; WP high = protected, bridge JP2 to program)

C10 100nF
U4 CAT24C32
R2 3.9k
R3 3.9k
R5 1k
JP2 EEPROM write enable

1-Wire (GPIO4 / XIAO D5 via JP6)

R1 4.7k
J2 1-Wire GND/DQ/3V3
R16 100
U8 SRV05-4
R13 100k
C15 1uF

UART (GPIO14/15 / XIAO D6/D7) + GPIO17: series R, ESD

J3 pin 1 = 3.3 V (JP1) or switched/fused 5 V (JP3), never both

JP1 J3 VCC = 3V3
JP3 J3 VCC = 5V
C12 10uF 25V
F1 0.5A PTC
Q1 AO3401A
Q2 AO3400A
R11 100k
R12 10k
R14 100k
R15 10k
C16 100nF
C31 1uF
R8 1k
R9 1k
R10 1k
J3 UART (e.g. PMS5003)

R33 100 I2S_LRCLK
R34 100 I2S_BCLK
R35 100 I2S_DIN
R36 100 BUZ_PWM
R37 100 BTN
R38 100 XDS_SEL
R39 100 UART_TX
R40 100 UART_RX
R41 100 I2C_SDA
R42 100 I2C_SCL
R43 100 LED_DATA

Test points and mounting holes

TP1 +5V
TP2 +3V3
TP3 +3V3_CO2
TP4 GND
TP5 EE_WP
TP6 I2C_SDA
TP7 I2C_SCL
MH1 M2.5
MH2 M2.5
MH3 M2.5
MH4 M2.5

Particulate-matter port: own LDO, JP4 (closed) = 3.3 V for SEN6x, or cut JP4 + bridge JP5 = 5 V for SEN5x

U9 AP2112K-3.3
C17 1uF
C18 1uF
JP4 PM VCC = 3V3
JP5 PM VCC = 5V

5 V ideal-diode OR (LM66100 x2)

J4 PM sensor (SEN6x/SEN5x)
R30 100k
C30 1uF
U15 LM66100
C19 22uF 25V
C20 100nF
R29 100k
C29 1uF
U14 LM66100

VOC/NOx SGP41 (0x59)

R28 4.7
U13 SGP41
C27 1uF

Expansion: Qwiic (3.3 V I2C) + GPIO pads (Pi only)

C28 1uF
J5 Qwiic / STEMMA QT
J6 GPIO expansion (DNP)

Button GPIO27 / XIAO D4

R19 10k
SW1 Button
C23 100nF

Status LED (GPIO10 / XIAO D10, 5 V via 74LV1T125) and buzzer (GPIO13 / XIAO D3, PWM)

C26 100nF
U12 74LV1T125
R32 100k
D1 WS2812B-2020
R20 100
C24 100nF
R21 33 0.5W 1206
LS1 MLT-5020 4kHz
D2 1N4148W
Q3 AO3400A
R22 100
R23 100k

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